Revenue Management

Pricing Algorithms

Agenda

1. Revenue management game

2. Fixed price heuristic

3. Two-price heuristic

Retailer Markdown Game

- Initial Stock: 2000 Units
  - Demand is difficult to predict.
  - No restocking.
- Initial Price: $60
  - Can markdown: $54 (10%), $48 (20%), $36 (40%).
- 15 week selling season
  - No salvage value.

GOAL?

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GOAL? Maximize revenues from the 2000 units.

Game Outline

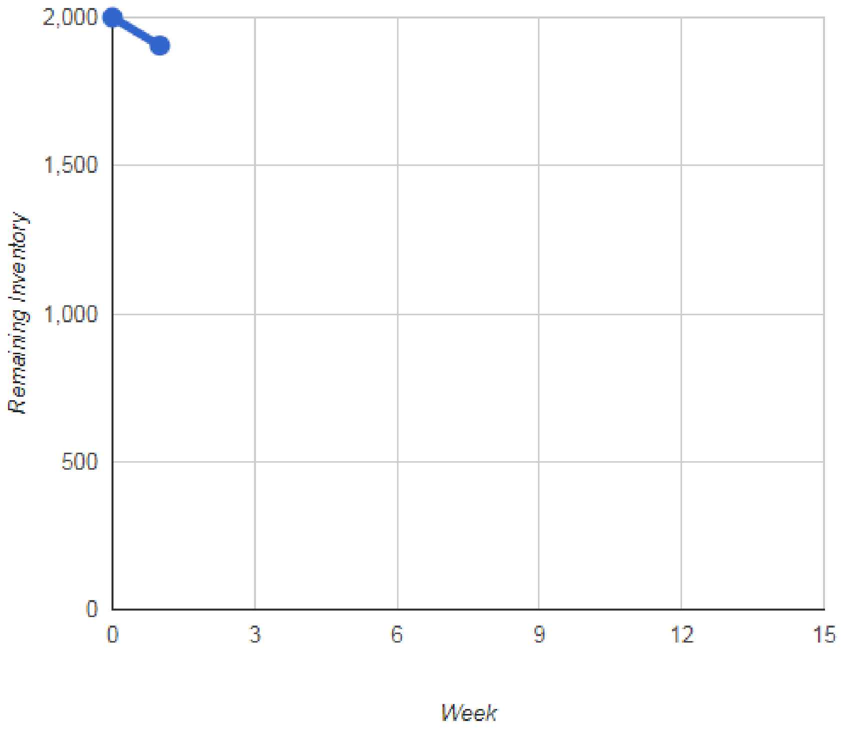
- Markdown Objective:
  - Develop a generic **markdown pricing strategy** for a retailer.
  - Maximize the total revenue when selling some inventory over a limited time period (i.e., at the end of week 15).

- Link to Game: http://www.randhawa.us/games/retailer/nyu.html

Interface – Columns

Restart Game **Compete Mode**

Week	Price	Sales	Remaining Inventory
1	60	95	1905



Maintain Price **10%** **20%** **40%**

Your revenue: $5,700

Columns Definition

- **Week** - Season goes from week 1 to week 15 (i.e., **14 decisions**).
- **Price ($)** - Price charged on a given week.
- **Sales** - Amount sold (not demand) on a given week.
- **Remaining Inventory** - Inventory on hand at the end of the week (beginning of the next week).

Playing the Game – Decisions

- Four possible price levels:
  - Initial price of $60.
  - 10% off ($54).
  - 20% off ($48).
  - 40% off ($36).

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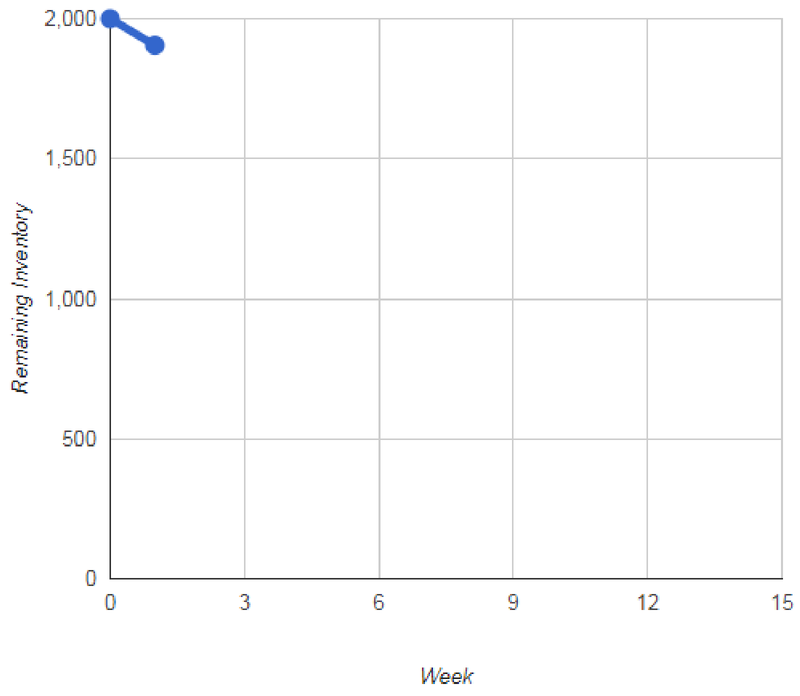
- Four possible price levels:
  - Initial price of $60.
  - 10% off ($54).
  - 20% off ($48).
  - 40% off ($36).
- Each week you have the option to maintain or decrease the price.
- **The price of the item cannot be raised.**
  Therefore, if at some point you choose 40% off, the game ends.

Interface – Decisions

Restart Game | Compete Mode

Week	Price	Sales	Remaining Inventory
1	60	95	1905



Maintain Price | 10% | 20% | 40%

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Playing the Game – Objective

- There are 15 weeks in the selling season:
  - 14 weeks for decisions, since the $60 price is set for the first week.

- The initial stock is 2000 units.

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- There are 15 weeks in the selling season:
  - 14 weeks for decisions, since the $60 price is set for the first week.

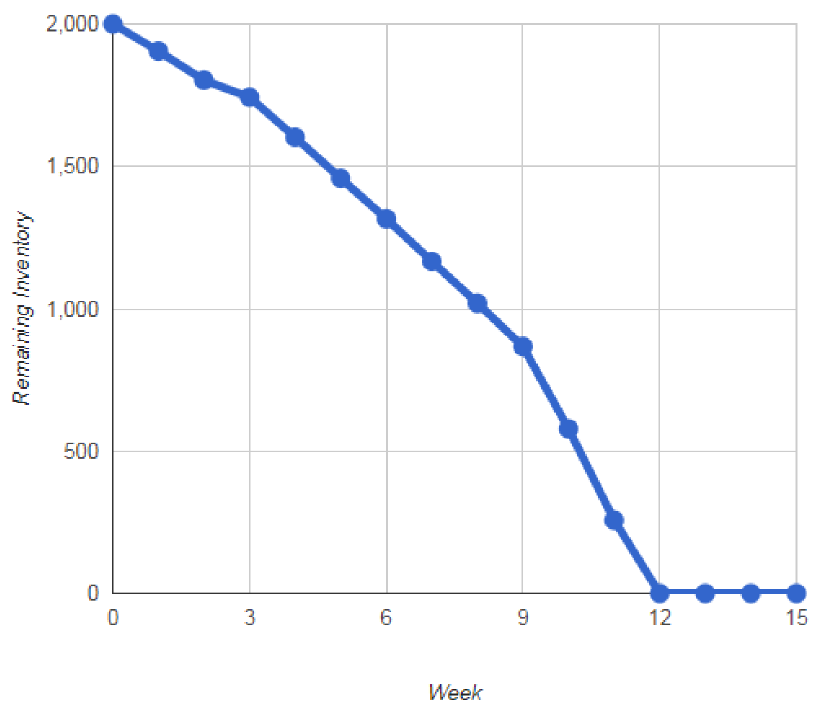
- The initial stock is 2000 units.

- The objective is to **maximize the total revenue** from the available stock during the 15 weeks.

Interface – Total Revenue

Restart Game Compete Mode

Week	Price	Sales	Remaining Inventory
1	60	95	1905
2	60	102	1803
3	60	60	1743
4	54	141	1602
5	54	144	1458
6	54	142	1316
7	54	150	1166
8	48	147	1019
9	48	152	867
10	48	289	578
11	36	321	257
12	36	257	0
13	36	0	0
14	36	0	0
15	36	0	0



Maintain Price 10% 20% 40%

Your revenue: $95,610, Perfect foresight strategy: $107,484, Difference: 11.0%

Playing the Game – No Costs

- The stock left-over at the end of the 15 weeks is lost. There is no salvage value.

- There are no costs to worry about since the production costs for the items are already sunk.

Interface – Printout (Example)

Restart Game | Compete Mode

Maintain Price

10%

Maintain Price

20%

Maintain Price

40%

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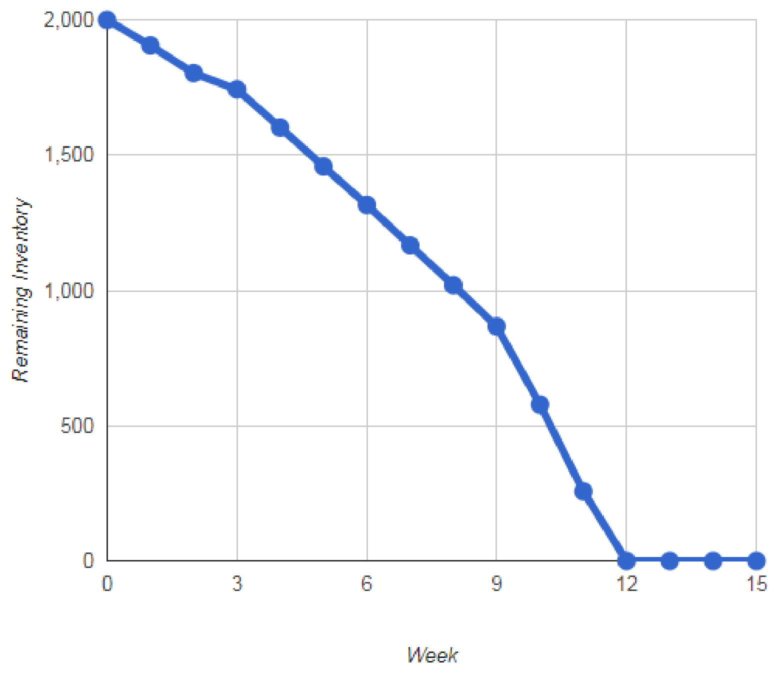

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Playing the Game

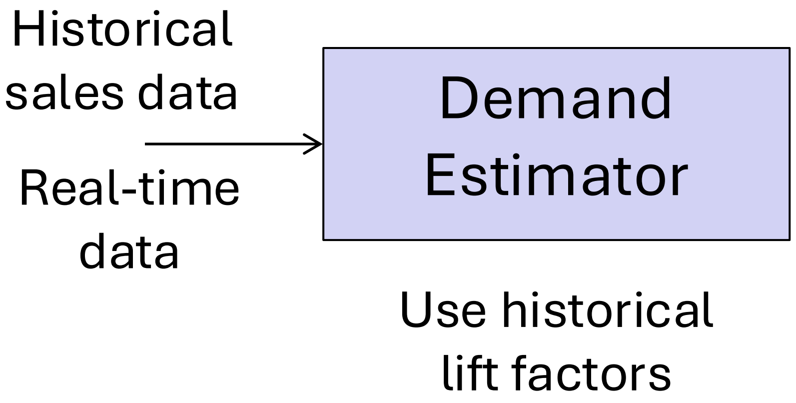
Class Discussion:

What can be a good strategy?

Markdown Strategy

<u>Requires two steps:</u>

- Demand estimation.
- Markdown strategy.



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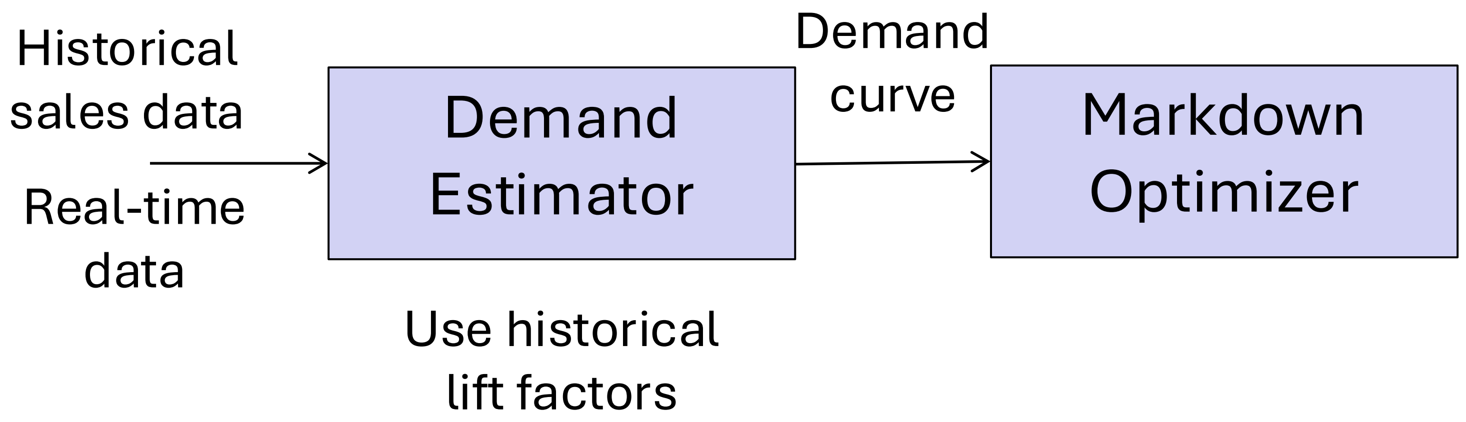

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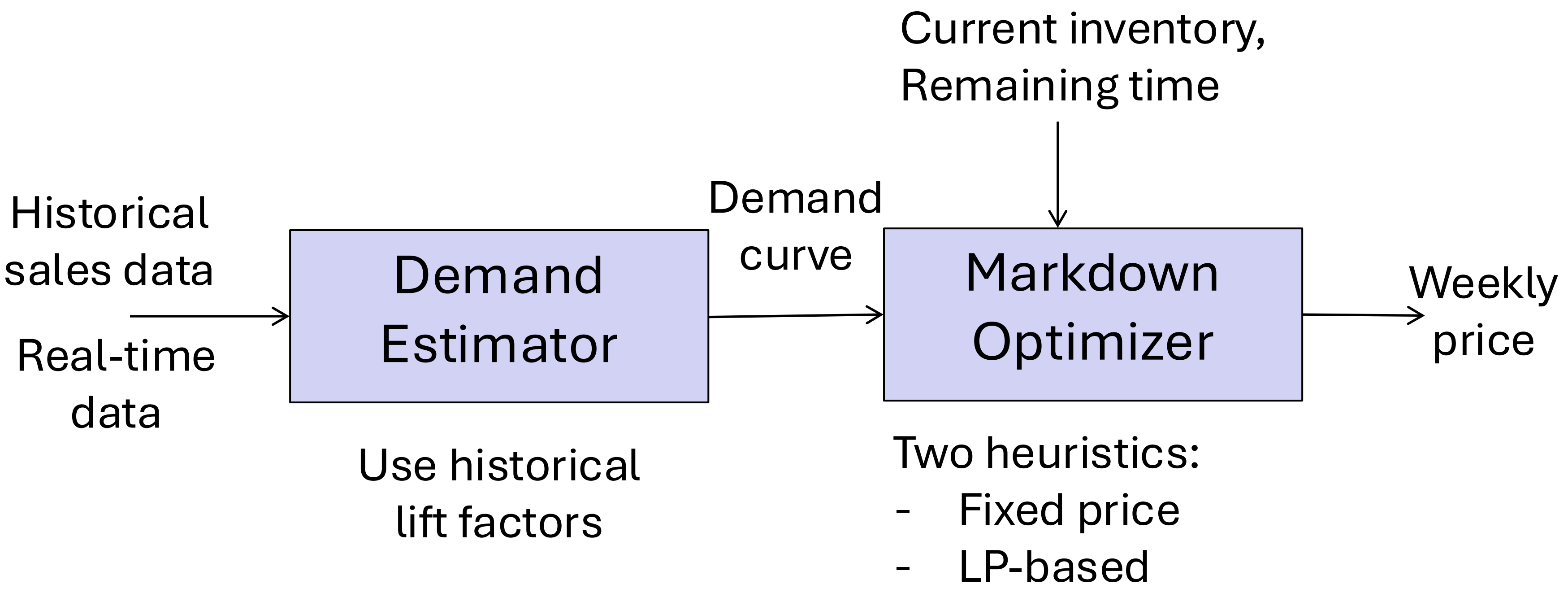
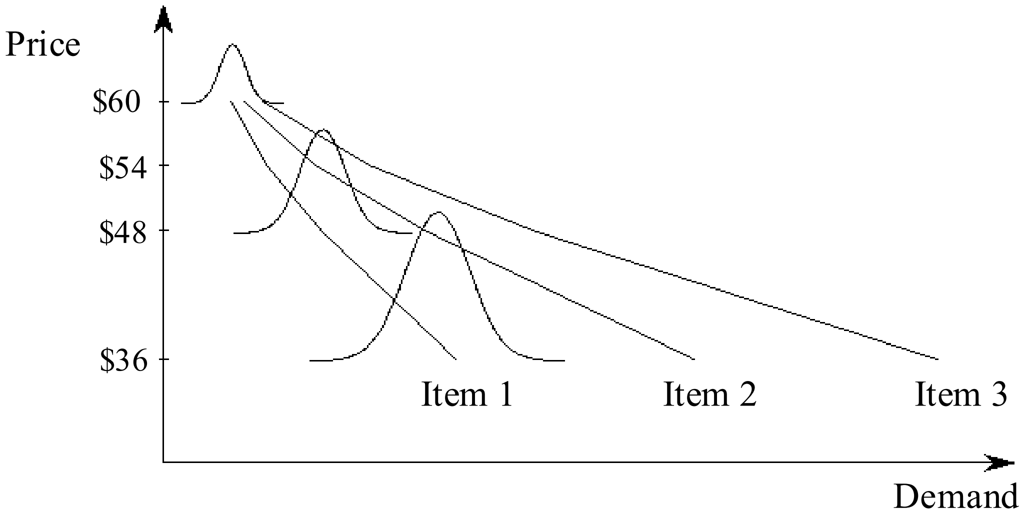


Demand Estimation



- Different demand curves for different items.
- Demand is random from week to week (even at the same price).
- Figuring out what curve you face:
  - Need historical data.
  - And real-time data.

Demand Estimation: Historical Data

Average weekly sales at a given price

Item	60	54	48	36
1	58	76		
2	108	144		
3	59	82		
4	61	78		
5	93	114		
6	114		209	
7	67		120	
8	53		97	
9	74		132	
10	67		97	
11	100			264
12	64			189
13	66			197
14	61			164
15	62			175

Bottom line: The absolute demand varies greatly from item to item but the <u>relative</u> lift from a price drop is consistent across items.

Demand Estimation: Real-Time Data

Say we observe sales of 59 units in week 1.

Then, we can assume:

- Sales of 59 units every week at $60.
- Sales of 59 x 1.31 = 77 units / week at $54.
- Sales of 59 x 1.73 = 102 units / week at $48.
- Sales of 59 x 2.81 = 166 units / week at $36.

Real-time data

Historical lift factor

The **lift factor** allows us to estimate the demand at different price levels

Revenue Prediction

- Full price demand = 59 units/week.

Price	Lift factor	Weekly demand (=full price demand x lift factor)	Weekly revenues (=Price x Weekly demand)
$60	1	59	$3,540
$54	1.31	77	$4,158
$48	1.73	102	$4,896
$36	2.81	166	$5,976

- A lot of time to sell 2000 units: sell at $60.
- Very little time to sell 2000 units: sell at $36.

Markdown Optimization: Fixed Price Heuristic

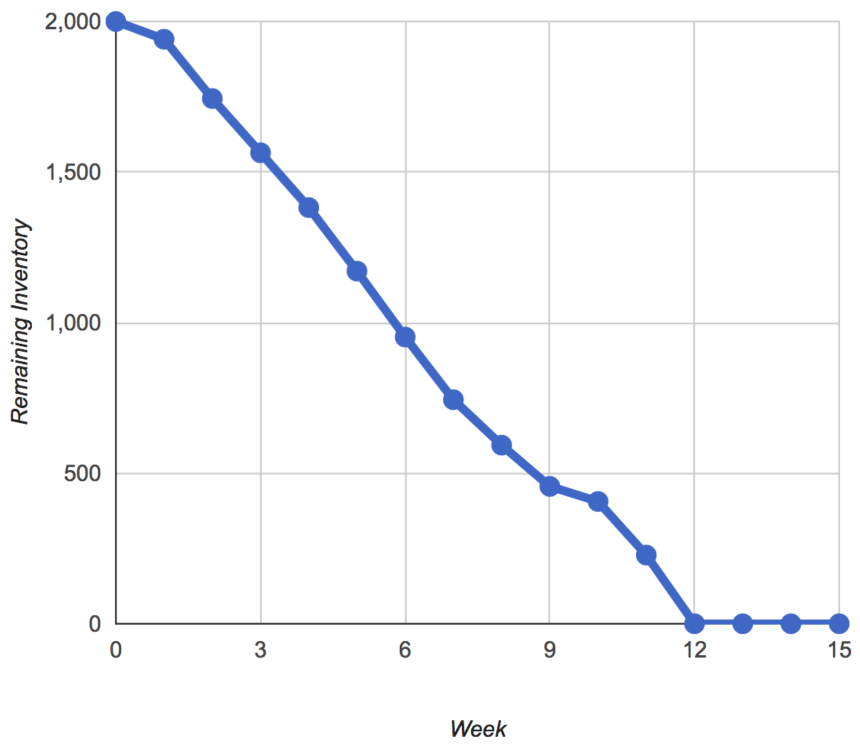
- Full price demand = 59 units/week.

Price	Lift factor	Weekly demand (=full price demand x lift factor)	Weekly revenues (=Price x Weekly demand)	Sales in remaining 14 weeks (=min(1941, Weekly demand x 14)	Revenues (=Sales x Price)
$60	1	59	$3,540	826	$49,560
$54	1.31	77	$4,158	1078	$58,212
$48	1.73	102	$4,896	1428	$68,544
$36	2.81	166	$5,976	1941	$69,876

- Fixed price heuristic:
  - Charge always the same price.
  - Best to **charge $36 every week** (from week 2 to week 15).
  - Total expected revenue = 3,540 + 69,876 = $73,416.

Fixed Price Heuristic

Week	Price	Sales	Remaining Inventory
1	60	59	1941
2	36	197	1744
3	36	180	1564
4	36	182	1382
5	36	211	1171
6	36	219	952
7	36	208	744
8	36	151	593
9	36	137	456
10	36	50	406
11	36	178	228
12	36	228	0
13	36	0	0
14	36	0	0
15	36	0	0



Maintain Price | 10% | 20% | 40%

Your revenue: $73,416, Perfect foresight strategy: $82,896, Difference: 11.4%

Markdown Optimization: Two-Price Heuristic

- <u>Two-price heuristic:</u> **1 week at \$48 and 13 weeks at \$36**.
  - Sales = min{1 x 102 + 13 x 166, 1941} = 1941.
  - Expected revenue = 3,540 + 1 x 48 x 102 + 1839 x 36 = \$74,640.

- How can we improve further?

- Ideally, <u>we want to clear the inventory exactly in the last week of the season</u>.

Markdown Optimization: Run-Out Rate Heuristic

Set the price combination such that we run out of inventory "**just in time**".

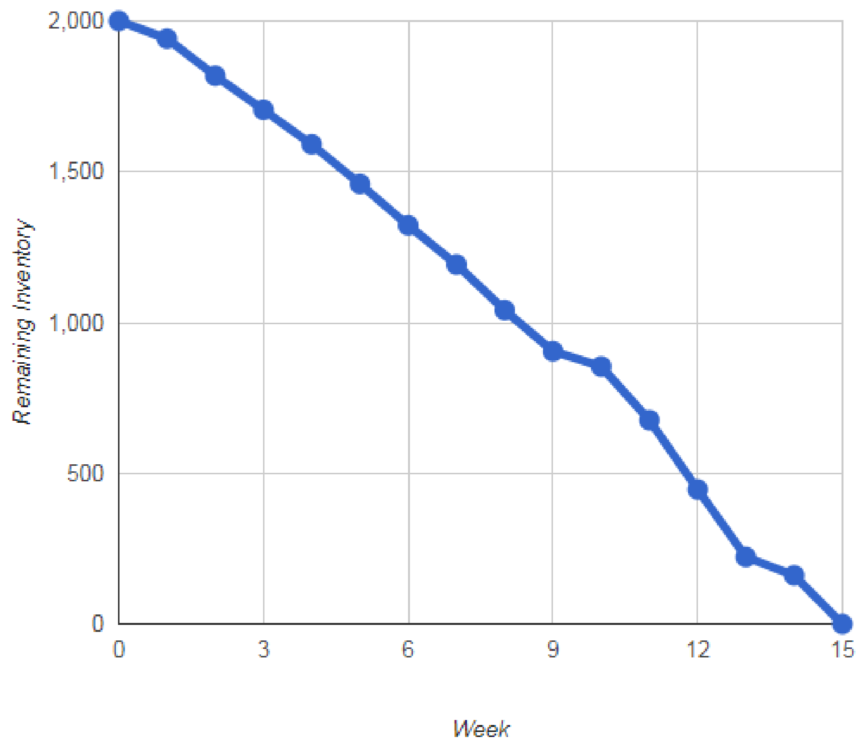
This is called the **run-out rate heuristic**.

Here using two prices:

Weeks at $48	2	3	4	5	6
Weeks at $36	12	11	10	9	8
Total demand (=Weeks at $48 x 102 + Weeks at $36 x 166)	2196	2132	2068	2004	**1940**
Units available	1941	1941	1941	1941	1941

Run-Out Rate Heuristic

Week	Price	Sales	Remaining Inventory
1	60	59	1941
2	48	123	1818
3	48	113	1705
4	48	114	1591
5	48	132	1459
6	48	137	1322
7	48	130	1192
8	36	151	1041
9	36	137	904
10	36	50	854
11	36	178	676
12	36	230	446
13	36	224	222
14	36	61	161
15	36	161	0



Maintain Price | 10% | 20% | 40%

Your revenue: $82,404, Perfect foresight strategy: $82,896, Difference: 0.6%

LP Heuristic - Decision Variables

We define the following decision variables:

x_{60} = # of weeks the item is sold at $60

x_{54} = # of weeks the item is sold at $54

x_{48} = # of weeks the item is sold at $48

x_{36} = # of weeks the item is sold at $36

LP Heuristic - Constraints

- Limited time:

$$x_{60} + x_{54} + x_{48} + x_{36} \leq 14$$

No. sale weeks

Season length
(remaining weeks)

- Limited inventory:

$$59{*}x_{60} + 77{*}x_{54} + 102{*}x_{48} + 166{*}x_{36} \leq 1941 = (2000 - 59)$$

Predicted sales

Inventory on hand

LP Heuristic – Formulation

Max Revenue

$$60*59*x_{60} + 54*77*x_{54} + 48*102*x_{48} + 36*166*x_{36}$$

Revenue from sales

s.t.

Limited time $\quad x_{60} + x_{54} + x_{48} + x_{36} \leq 14$

Limited inventory $\quad 59*x_{60} + 77*x_{54} + 102*x_{48} + 166*x_{36} \leq 1941$

Excel Output

Retailer Linear Programming Worksheet

								Crucial Input Parameters
Weekly demand at full price		59	<- change this value and re-solve the LP					Decision Variables
Number of weeks remaining		14						Objective
Initial quantity on hand		1941						
Salvage value		0						
Price levels	60	54	48	36				
Demand multipliers	1	1.31	1.73	2.81				
Avg. weekly demand	59	77.3	102.1	165.8				
					Total	**Constraint**		
No. of weeks at price	0.0	0.0	6.0	8.0	14.0	<=	14	
Total sold at price	0.0	0.0	608.8	1332.2	1941	<=	1941	
Revenue Computations:								
Revenue from sales	77,182							
Revenue from salvage	0							
Total revenue	$77,182							

Total expected revenue including week 1: $3,540 + $77,182 = $80,772

Plan: Weeks 2-7 @ 20%, Weeks 8-15 @ 40%.

LP Solution

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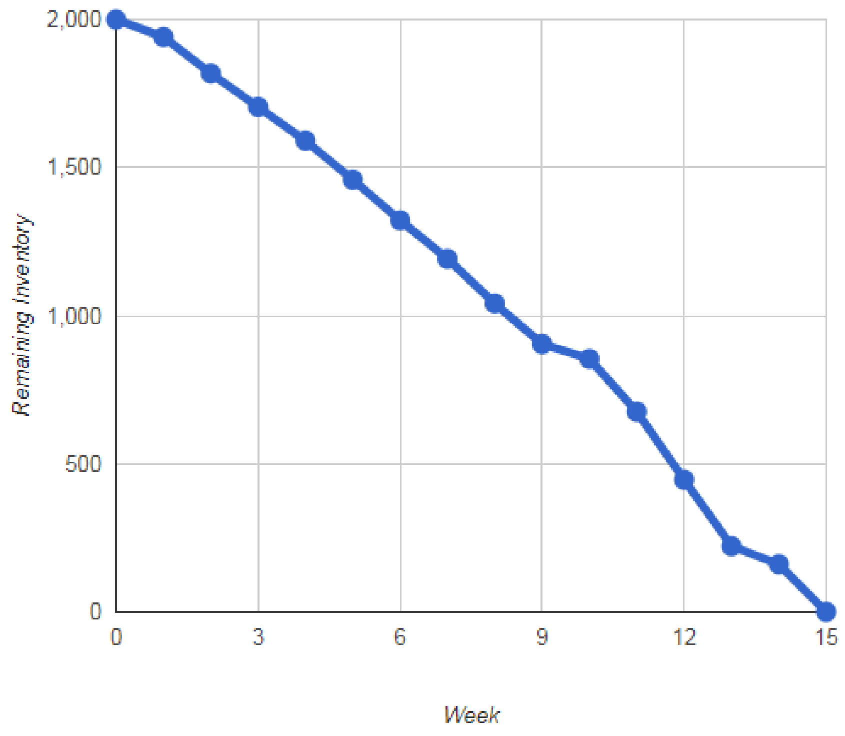
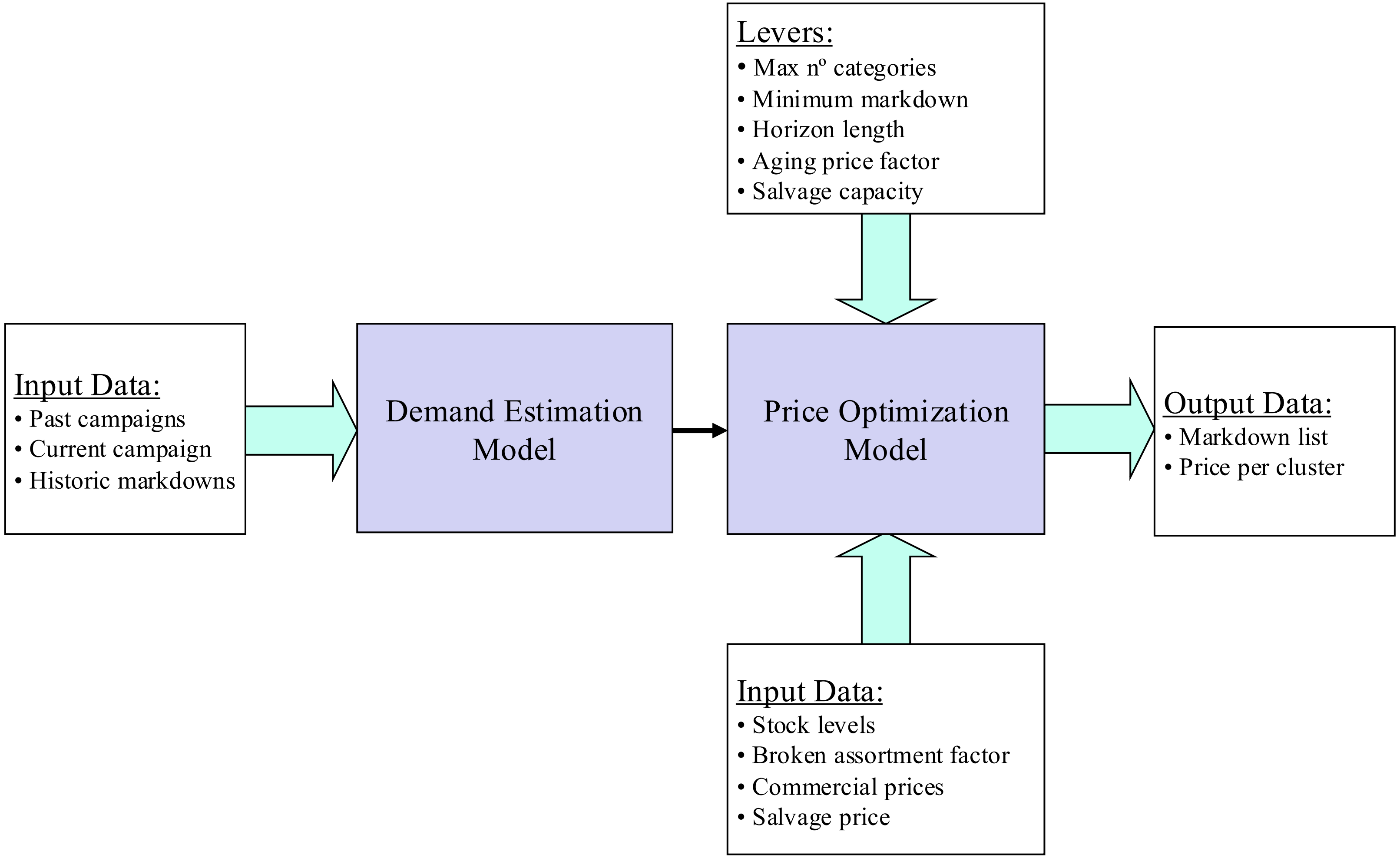


Maintain Price | 10% | 20% | 40%

Your revenue: $82,404, Perfect foresight strategy: $82,896, Difference: 0.6%

(Same as the run-out rate heuristic.)

Pricing Process at Zara



These model and results are based on the work of Felipe Caro and Jeremie Gallien.